Quadratic Expressions and Equations

Worksheet 2 (Calculator)

1. W13/qp21/q5(a,b,c)

a. Solve $\frac{2}{3-x} = 1$. [1]

b. Factorise

i. $5x + 5y$ [1]

ii. $9x^2 - 16$ [1]

c.

i. Factorise $2x^2 + 5x - 12$. [1]

ii. Use your answer to part (c)(i) to solve the equation $2x^2 + 5x - 12 = 0$. [1]

2. S17/qp21/q7(a,b)

a. Factorise completely $12a^2b - 15ab^3$. [1]

b.

i. Write $4x^2 + 12x + 9$ in the form $(cx + d)^2$. [1]

ii. Hence solve $4x^2 + 12x + 9 = 49$. [2]

3. S17/qp22/q5(b,c)

a. Simplify $4(3x - 2y + 1) - (5x - 3y + 1)$. [2]

b. Solve $3x^2 - x - 5 = 0$, giving your answers correct to 2 decimal places. [3]

4. S18/qp21/q3(c)

Simplify $\frac{m^2+3m}{2m^2+5m-3}$. [3]

5. S18/qp22/q3(c)

Simplify $\frac{v^2-8v}{2v^2-13v-24}$. [3]

6. S19/qp21/q4

a. Express as a fraction in its simplest form.

i. $\frac{6y}{35} \div \frac{10y^2}{7}$ [2]

ii. $\frac{k^2-16}{k^2-2k-8}$ [3]

b. Solve $3(x-4)+5=7$. [2]

c. Solve $3t^2 + 5t - 4 = 0$. Show all your working and give your answers correct to 2 decimal places. [3]

Answer Key:

1.(a)1 (b)(i) 5(x+y) (ii) (3x+4)(3x-4) (c)(i)(2x-3)(x+4) (ii) $\frac{3}{2}$ −4 (d)4

2.(a)$3ab(4a - 5b^2)$ (b)(i) $(2x + 3)^2$ (ii)2,-5

3.(b) 7x-5y+3 (c) -1.14, 1.47

4. $\frac{m}{2m-1}$

5. $\frac{v}{2v+3}$

6. (a)(i)$\frac{3}{25y}$ (ii) $\frac{k+4}{k+2}$ (b) $\frac{14}{3}$ (c)0.59 and -2.26